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标题	Cascade Iodine-Catalyzed Synthesis of Nitrogenated Aromatics from 2 - Pyrones
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摘要（原文节选）

Herein, we report an improved strategy for the synthesis of nitrogen-containing aromatic compounds featuring a nitrogen atom directly attached to a benzene ring. The method relies on a thermal one-pot Diels – Alder/decarboxylation/iodine-catalyzed aromatization cascade reaction starting from 2-pyrones, enabling access to novel C – 3-nitrogenated phthalimides and related aromatics in yields of up to 95%. Iodine, a simple and readily available catalyst, was employed to promote the transformation while effectively suppres ⋯

数据表 · 1（14 行，含数值行 14 行）

Screening of Reaction Parameters for the Diels-Alder reaction/Decarboxylation/Aromatization Steps

entry	equiv NEM	equiv 2	time (h)	yield 2a (%)	yield 3a (%)
1	3.0	none	15	32	38
2	3.0	1.0	4	83	-
3	3.0	0.5	4	84	-
4	3.0	0.25	4	85	-
5	3.0	0.1	4	82	-
6	3.0	0.05	4	53	22
7	2.0	0.1	4	60	-
8	2.0	0.1	15	92 (84)	-
9	1.0	0.1	15	50	-
10	2.0	0.1	2	76	-
11	2.0	0.1	4	77	-
12	3.0	0.1	2	83	-
13	2.0	0.1	15	77	-
14	2.0	0.1	15	69	-

脚注（逐字）：aYields and conversion were determined by 1H NMR analysis using 1,3,5-Trimethoxybenzene as an internal standard.bIsolated yield.cReaction performed under microwave irradiation.dReaction performed at 175 ° C.eReaction performed at 150 ° C.

正文对应句（逐字）：Next, this methodology was further extended to a variety of N-substituted maleimides, as well as other classical dienophiles (for dienophiles that did not afford the product, see the Supporting Information). Overall, N-alkyl maleimides resulted in compounds 2i – k in lower yields when comparing to N-a

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